

Effects of Na₂O and BaO Additions on Phosphorus Distribution between CaO–MgO–Fe_tO–SiO₂ Slags and Liquid Iron*

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Synopsis

Experiments have been made to study the effect of Na₂O (7~13 wt%) and BaO (4 wt%) additions on the phosphorus distribution between liquid iron and CaO–MgO–Fe_tO–SiO₂–P₂O₅ (≤1 wt%) slags at 1 550 °C. The effect of Na₂O and BaO on the logarithm of the equilibrium quotient, $k_p (= (\%P_2O_5)/([\%P]^2(\%Fe_tO)^5)$, was found to be 1.2 and 0.9 times as large as that of CaO on weight percent basis, respectively. The values for $\log k_p$ can be expressed as

$$\log k_p = 0.145[(\%CaO) + 0.3(\%MgO) - 0.5(\%P_2O_5) + 1.2(\%Na_2O) + 0.9(\%BaO)] + 22\,810/T - 20.506$$

The effect of Na₂O or BaO on the activity coefficients of P₂O₅ was also studied by the Turkdogan and Pearson's treatment. The following expression for the $\log \gamma_{P_2O_5}$ is obtained for the Na₂O- or BaO-containing slags.

$$\log \gamma_{P_2O_5} = -1.02(23N_{CaO} + 17N_{MgO} + 8N_{Fe_tO} - 26N_{P_2O_5} + 33N_{Na_2O} + 42N_{BaO}) - 22\,990/T + 9.490$$

It was found from the plot of activity coefficients of Fe_tO vs. mole fraction of SiO₂ that activity coefficients of Fe_tO decrease with the addition of Na₂O, while it does not change with the addition of BaO. The dephosphorization equilibrium was discussed on the basis of the theoretical optical basicity proposed by Duffy and Ingram. The following equation was obtained as a function of optical basicity.

$$\log N_{PO_4^{3-}}/([P][O]^{5/2}) = 22.22A_{th} + 30\,890/T - 28.752$$

where, $N_{PO_4^{3-}}$: the anionic fraction of PO₄³⁻ ion

[]: weight percent in metal phase.

I. Introduction

The studies on the phosphorus distribution between liquid iron and the flux containing very basic oxides such as Na₂O, BaO, etc. are important for the production of the steel with extremely low content of phosphorus. The present authors have previously studied the phosphorus distribution between liquid iron and MgO-saturated CaO–Fe_tO–SiO₂–P₂O₅ (≤1 wt%) slags in the temperature range of 1 550 to 1 650 °C.¹⁾ Subsequently, the effect of CaF₂ (1~4 wt%) addition of the phosphorus distribution between liquid iron and above-mentioned slags has been studied.²⁾ Phosphorus distribution ratios between liquid iron and CaO–MgO_{sat.}–Fe_tO–SiO₂–P₂O₅ slags containing 2~5 wt% MnO have also been obtained^{3,4)} and on the basis of these results, it is demonstrated that the equilibrium quotient for the dephosphorization reaction can be approximated in terms of a linear function of slag compositions.

The purpose of this investigation is to examine the effects of Na₂O (7~13 wt%) and BaO (4 wt%) additions on the phosphorus distribution ratios by the same method employed in the previous works.¹⁻⁴⁾ Recently, Gaskell⁵⁾ studies the relationship between

optical basicity and the dephosphorization equilibrium by use of our data¹⁾ and then derives the dephosphorization equation as a function of optical basicity. The present and previous results^{1,2,4)} for the dephosphorization equilibrium are also discussed on the basis of optical basicity.

II. Experimental

The experimental apparatus and procedure are the same as those described elsewhere.¹⁾ Premelted Na₂O·SiO₂ and reagent grade BaO were used for the additions. The equilibrium of phosphorus was approached mostly from the slag side, but in some experiments from the metal side by adding an Fe–26%P alloy. The P₂O₅ content in the final slags was controlled to be less than 1 %. All the experiments were carried out at 1 550 °C only. The holding time was chosen from 30 to 60 min in the experiments with Na₂O, since in the preliminary experiments significant evaporation loss of sodium was observed in some compositions of slags containing Na₂O. In the experiments with BaO, the holding time was from 3.5 to 5 hr depending on the slag compositions. The degree of the evaporation loss of sodium from the total sodium oxide added was found to be about 0~24 %. About half of MgO in the MgO-saturated slags determined in the previous experiments¹⁾ was from MgO addition to the slag prior to melting. This MgO content is considered to be under-saturated because of the short holding time. The contents of CaO, SiO₂, MgO, Fe²⁺, total Fe and metallic Fe in slags and P and Q in metal are analyzed by the method described elsewhere.^{2,6)} Sodium was determined by the atomic absorption spectrochemical analysis (JIS-M-8204) and BaO was determined by BaSO₄ gravimetric method after dissolving in HCl.^{7,8)}

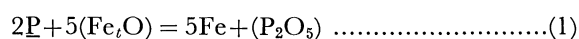
III. Results

The compositions of the equilibrated slag and metal are summarized in Tables 1 and 2.

IV. Discussion

1. Representation of Phosphorus Equilibrium

It was found in the previous investigation⁴⁾ that the logarithms of the equilibrium quotient k_p , Eq. (2), for the dephosphorization reaction, Eq. (1), can be expressed by Eq. (3).



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Table 1. Equilibrium slag and metal compositions.
CaO-MgO_{sat.}-FeO_x-SiO₂-BaO (1 550 °C)

No.	P	O	FeO	Fe ₂ O ₃	CaO	MgO	SiO ₂	P ₂ O ₅	BaO
801	0.0084	0.0996	21.68	2.20	27.12	14.20	29.42	0.726	3.98
802	0.0109	0.1130	33.14	2.79	19.02	14.84	24.77	0.707	4.19
803	0.0036	0.1121	28.95	4.69	30.05	9.30	21.96	0.858	4.12
804	0.0067	0.1153	41.99	4.89	20.30	9.83	17.42	0.783	4.16
805	0.0026	0.1201	38.22	6.98	29.21	6.23	14.14	0.803	4.14
806	0.0032	0.1250	44.64	10.44	24.61	5.80	9.29	0.817	4.06
807	0.0128	0.1404	58.06	7.29	12.78	7.06	9.32	0.758	4.26
808	0.0067	0.1260	60.65	12.18	16.26	5.10	0.50	0.854	3.72
809	0.103	0.1876	80.14	8.38	0.19	6.02	0.59	0.309	3.40
810	0.0858	0.1690	68.95	5.72	0.19	12.15	8.24	0.409	3.57
811	0.0687	0.1463	58.25	4.39	0.29	17.81	15.11	0.422	3.83
812	0.0074	0.1259	45.92	5.82	18.99	8.85	15.18	0.801	3.99

Table 2. Equilibrium slag and metal compositions.
CaO-MgO_{sat.}-FeO_x-SiO₂-Na₂O (1 550 °C)

No.	P	O	FeO	Fe ₂ O ₃	CaO	MgO	SiO ₂	P ₂ O ₅	Na ₂ O
701	0.0039	0.0851	23.56	4.23	25.61	10.05	28.10	0.764	8.82
702	0.0063	0.0934	29.08	4.46	18.38	10.30	26.60	1.27	11.2
703	0.0040	0.0900	26.21	5.53	29.86	6.93	22.15	1.40	8.37
705	0.0016	0.1025	38.93	8.56	26.93	5.07	13.02	0.800	7.24
706	0.0037	0.1116	35.16	6.73	17.28	8.50	21.67	0.963	10.5
707	0.0063	0.1252	42.26	5.92	9.61	9.45	21.38	1.01	11.5
709	0.0016	0.1043	32.23	7.23	28.30	6.26	17.82	0.856	7.99
710	0.0031	0.1190	37.65	7.42	20.83	6.69	17.05	1.15	9.48
711	0.0026	0.1214	43.83	8.21	13.03	6.66	16.23	0.761	10.9
712	0.0033	0.1289	49.88	8.05	6.65	6.38	15.11	0.753	12.8
713	0.0016	0.1242	45.36	8.99	16.52	5.28	12.47	0.764	11.0

$k_p = (\%P_2O_5)/[(\%P)^2(\%Fe_tO)^5] \dots\dots\dots (2)$
 $\log k_p = 0.145[(\%CaO)+0.3(\%MgO) - 0.5(\%P_2O_5)] + 22\,810/T - 20.506 \dots\dots (3)$

The values for $\log k_p$ calculated from the previous data¹⁾ for the CaO-MgO_{sat.}-Fe_tO-SiO₂-P₂O₅ slags are plotted against the $[(\%CaO)+0.3(\%MgO)-0.5(\%P_2O_5)]$ term by small solid circles in Fig. 1, where the line corresponds to Eq. (3) at 1 550 °C. Logarithms of k_p obtained in the experiments containing 7~13 wt% Na₂O and 4 wt% BaO are plotted against the $[(\%CaO)+0.3(\%MgO)-0.5(\%P_2O_5)]$ by large solid circles and triangles, respectively. To analyze the CaO-equivalency of Na₂O or BaO graphically, the difference, $\Delta \log k_p$, between the line and the data points indicated by solid circles or triangles are plotted as a function of the content of Na₂O or BaO. From the slope of this plot, the CaO-equivalencies of Na₂O and BaO were obtained as 1.2 ± 0.2 and 0.9 ± 0.2 , respectively on weight percent basis. The more details of derivation of the above constant are described elsewhere.²⁾ The values for $\log k_p$ indicated by open circles and triangles are plotted against the $[(\%CaO)+0.3(\%MgO)-0.5(\%P_2O_5)+1.2(\%Na_2O)+0.9(\%BaO)]$ term. The difference between open and solid circles at the constant $\log k_p$ corresponds to

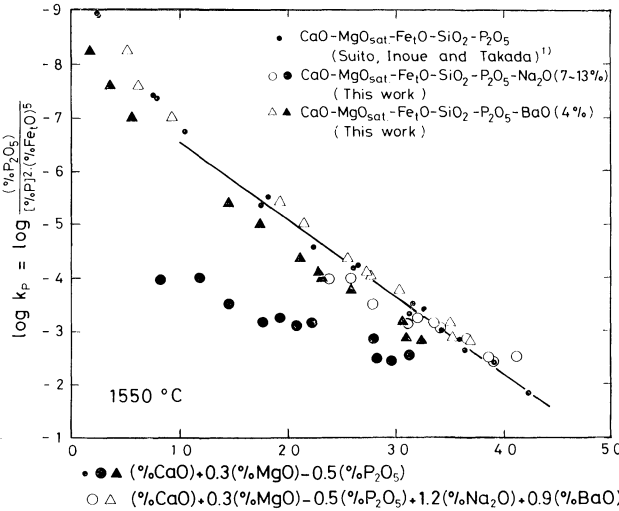


Fig. 1. Logarithms of the equilibrium quotient k_p as a function of $[(\%CaO)+0.3(\%MgO)-0.5(\%P_2O_5)+1.2(\%Na_2O)+0.9(\%BaO)]$ for Na₂O- or BaO-containing and -free slags.

the $1.2(\%Na_2O)$; between open and solid triangles, the $0.9(\%BaO)$. These open circles and triangles are fairly well on the line, Eq. (3), obtained from the data indicated by small solid circles.

Balajiva *et al.*'s relationship⁹⁻¹¹⁾ obtained in the previous data^{1,4)} on the $\text{CaO-MgO}_{\text{sat}}\text{-Fe}_t\text{O-SiO}_2\text{-P}_2\text{O}_5$ slags is given as⁴⁾

$$\begin{aligned} \log k_p = & 7.87 \log [(\% \text{CaO}) + 0.3(\% \text{MgO}) \\ & - 0.05(\% \text{Fe}_t\text{O}) - 0.5(\% \text{P}_2\text{O}_5)] \\ & + 22\,240/T - 27.124 \quad \dots\dots\dots(4) \end{aligned}$$

The values for $\log k_p$ obtained from the experiments containing Na_2O or BaO are plotted against the $\log [(\% \text{CaO}) + 0.3(\% \text{MgO}) - 0.05(\% \text{Fe}_t\text{O}) - 0.5(\% \text{P}_2\text{O}_5)]$ term by solid circles and triangles in Fig. 2. The line corresponding to Eq. (4) at 1550°C was obtained from the data for the Na_2O - and BaO -free slags. Logarithms of k_p indicated by open circles and triangles are plotted against the $\log [(\% \text{CaO}) + 0.3(\% \text{MgO}) - 0.05(\% \text{Fe}_t\text{O}) - 0.5(\% \text{P}_2\text{O}_5) + 1.2(\% \text{Na}_2\text{O}) + 0.9(\% \text{BaO})]$ term. It may be seen that these points lie well on the line. Usui *et al.*¹²⁾ obtained the coefficient of Na_2O in Eq. (4) as 1.58 from the results for the phosphorus distribution between liquid iron and MgO -saturated $\text{CaO-Fe}_t\text{O-SiO}_2$ slags at 1600°C , whose Na_2O content was higher than that in our study.

Logarithms of the $(\% \text{P})/([\% \text{P}](\% \text{Fe}_t\text{O})^{5/2})$ term are obtained by Healy¹³⁾ as a function of (wt% CaO) and temperature using ionic concept. This term was derived by using the previous data^{1,4)} as follows⁴⁾:

$$\begin{aligned} \log (\% \text{P})/([\% \text{P}](\% \text{T.Fe})^{5/2}) = & 0.0720(\% \text{CaO}) \\ & + 0.3(\% \text{MgO}) + 0.6(\% \text{P}_2\text{O}_5) \\ & + 11\,570/T - 10.520 \quad \dots\dots\dots(5) \end{aligned}$$

The values for $\log (\% \text{P})/([\% \text{P}](\% \text{T.Fe})^{5/2})$ obtained from the data for the slags with Na_2O or BaO are plotted against the $[(\% \text{CaO}) + 0.3(\% \text{MgO}) + 0.6(\% \text{P}_2\text{O}_5)]$ term as small solid circles in Fig. 3, where the line corresponds to Eq. (5) at 1550°C . The values for $\log (\% \text{P})/([\% \text{P}](\% \text{T.Fe})^{5/2})$ in the slags containing Na_2O or BaO are plotted against the $[(\% \text{CaO}) + 0.3(\% \text{MgO}) + 0.6(\% \text{P}_2\text{O}_5)]$ term as solid circles and triangles, respectively. The effect of Na_2O or BaO addition on the $\log (\% \text{P})/([\% \text{P}](\% \text{T.Fe})^{5/2})$ was studied by the same method described previously: the CaO -equivalencies of Na_2O and BaO were found to be 1.3 ± 0.2 and 0.9 ± 0.1 , respectively on weight percent basis. These values are also plotted against the $[(\% \text{CaO}) + 0.3(\% \text{MgO}) + 0.6(\% \text{P}_2\text{O}_5) + 1.3(\% \text{Na}_2\text{O}) + 0.9(\% \text{BaO})]$ term as open circles and triangles, respectively. These experimental points lie on the line for the data points given by small solid circles within the experimental accuracy.

The effect of CaF_2 (1~4 wt%) on the phosphorus distribution ratios between liquid iron and MgO -saturated $\text{CaO-Fe}_t\text{O-SiO}_2\text{-P}_2\text{O}_5$ (≤ 1 wt%) slags has been previously studied.²⁾ The relationships as shown in Figs. 1 and 2 have given in the previous reports.²⁾ However, FeO was used to represent the dephosphorization equations such as Eqs. (3) and (4). In the subsequent paper,⁴⁾ Eqs. (3) and (4) were revised by using Fe_tO as an iron oxide instead of FeO and by considering the effect of the P_2O_5 content on the

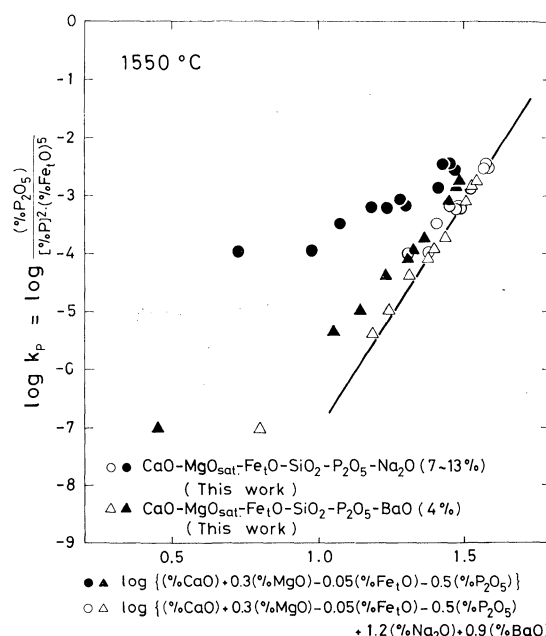


Fig. 2. Revised Balajiva *et al.*'s relationship⁹⁻¹¹⁾ for Na_2O - or BaO -containing slags.

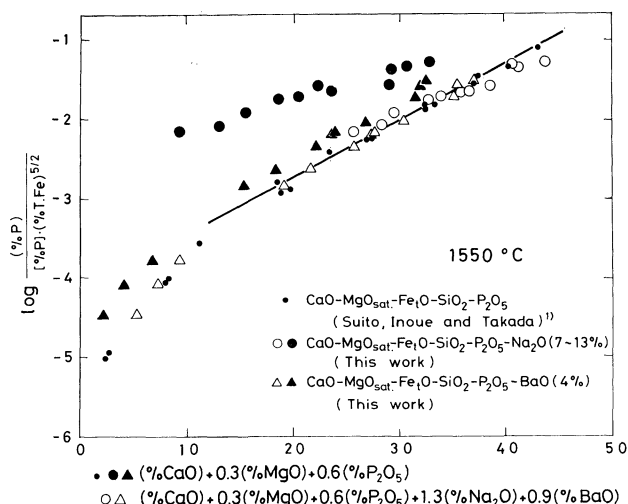


Fig. 3. Revised Healy's relationship¹³⁾ for Na_2O - or BaO -containing and -free slags.

dephosphorization equilibrium. Logarithms of $(\% \text{P})/([\% \text{P}](\% \text{T.Fe})^{5/2})$ in the CaF_2 -containing slags are plotted against the $[(\% \text{CaO}) + 0.3(\% \text{MgO}) + 0.6(\% \text{P}_2\text{O}_5)]$ term in Fig. 4 as solid circles. The coefficient of $(\% \text{CaF}_2)$ in the first term of the right hand side of Eq. (5) was obtained as 1.2 ± 0.3 by the same method described previously. This indicates that the CaO -equivalency of CaF_2 is 1.2 on weight percent basis. The values for $\log (\% \text{P})/([\% \text{P}](\% \text{T.Fe})^{5/2})$ are also plotted against the $[(\% \text{CaO}) + 0.3(\% \text{MgO}) + 0.6(\% \text{P}_2\text{O}_5) + 1.2(\% \text{CaF}_2)]$ term in Fig. 4 as small open circles which are well on the line given by Eq. (5) obtained from the data for the slags without CaF_2 .

2. Effects of Na_2O and BaO on Activity Coefficients of P_2O_5

In order to study the effect of Na_2O or BaO on the activity coefficient of P_2O_5 , the following Turk-

dogan and Pearson's relationship¹⁴⁾ obtained in the previous work has been employed.⁴⁾

$$\begin{aligned} \log \gamma_{P_2O_5} = & -1.02(23N_{CaO} + 17N_{MgO} + 8N_{Fe_tO} \\ & - 26N_{P_2O_5} + 13N_{MnO}) - 22\,990/T \\ & + 9.490 \dots\dots\dots(6) \end{aligned}$$

Logarithms of $\gamma_{P_2O_5}$ calculated from the previous data for the slags without Na₂O and BaO are plotted against the $(23N_{CaO} + 17N_{MgO} + 8N_{Fe_tO} - 26N_{P_2O_5})$ term as small solid circles in Fig. 5, where the line calculated from Eq. (6) at 1 550 °C is also indicated. The values for $\log \gamma_{P_2O_5}$ in the slags containing Na₂O or BaO are plotted against the $(23N_{CaO} + 17N_{MgO} + 8N_{Fe_tO} - 26N_{P_2O_5})$ term as solid circles and triangles, respectively. To analyze the CaO-equivalencies of Na₂O and BaO, the difference, $\Delta \log \gamma_{P_2O_5}$, between these points and lines was calculated. This difference divided by -1.02 , which is the coefficient of the first term of the right hand side of Eq. (6), was plotted against mole fractions of Na₂O or BaO: from the slope obtained from considerably scattering points, the coefficients of N_{Na_2O} and N_{BaO} were obtained as 33 ± 3 and 42 ± 12 , (leading thus to the CaO-equivalencies as 33/23 and 42/23), respectively. Logarithms of $\gamma_{P_2O_5}$ for the slags containing Na₂O or BaO are plotted against the $(23N_{CaO} + 17N_{MgO} + 8N_{Fe_tO} - 26N_{P_2O_5} + 33N_{Na_2O} + 42N_{BaO})$ term as open circles and triangles, respectively in Fig. 5. These points scatter along the line and are within the accuracy of the data points given by small solid circles. The value of coefficient of N_{CaF_2} in the previous report was 21, but the value of 20 is used in Eq. (6), since the recalculation of Turkdogan and Pearson's treatment was made using Fe_tO instead of FeO.

3. Effects of Na₂O and BaO on Activity Coefficients of Fe_tO

The dependence of activity coefficient of Fe_tO in the CaO-MgO_{sat}-Fe_tO-SiO₂-P₂O₅ (≤ 1 wt%) slags on the slag compositions has been previously studied by the method of multiple regression.⁴⁾ It was found that logarithms of γ_{Fe_tO} only depend on mole fraction of SiO₂. These values for $\log \gamma_{Fe_tO}$ at 1 550 °C are plotted against N_{SiO_2} in Fig. 6 as small solid circles which are fitted to the lines by the method of a quadratic form. The values for $\log \gamma_{Fe_tO}$ obtained from the present data on the slags containing Na₂O (7~13 wt%) and BaO(4 wt%) are plotted against N_{SiO_2} by the symbols ○ and ⊙, respectively. These results indicate that activity coefficients of Fe_tO decreases with the addition of Na₂O, while in case of the BaO addition it does not change.

4. Phosphate Capacity

Phosphate capacity, C_P is defined as

$$C_P \equiv (\%P)/(a_P a_O^{5/2}) \dots\dots\dots(7)$$

The value of C_P corresponds to the equilibrium quotient for the following dephosphorization reaction given by Eq. (8) and is proportional to the activity

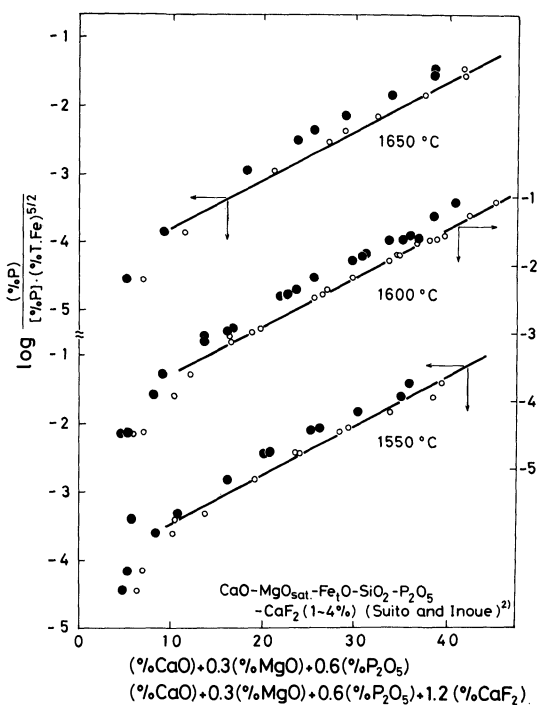


Fig. 4. Revised Healy's relationship¹³⁾ for CaF₂-containing slags.

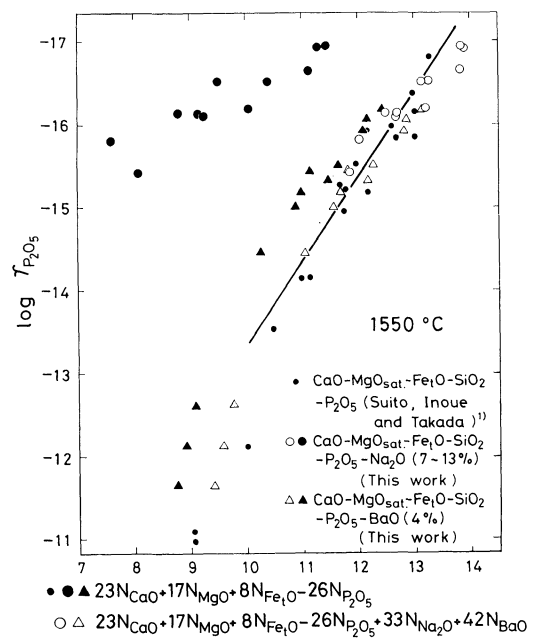


Fig. 5. Turkdogan and Pearson's relationship¹⁴⁾ for Na₂O- or BaO-containing and -free slags.

coefficient of PO_{2.5} at a constant temperature.

$$P + 5/2O = (PO_{2.5}) \dots\dots\dots(8)$$

$$\log C_P \{ \equiv \log (\%P)/(a_P a_O^{5/2}) \} \propto \log \gamma_{PO_{2.5}} \dots\dots\dots(9)$$

The phosphate capacity defined in Eq. (7) is a function of slag compositions and temperature: it is similar to the sulfide capacity defined by Richardson and Fincham.¹⁵⁾

Using the standard free-energy change ΔG° for the reaction given by Eq. (10), the following relation-

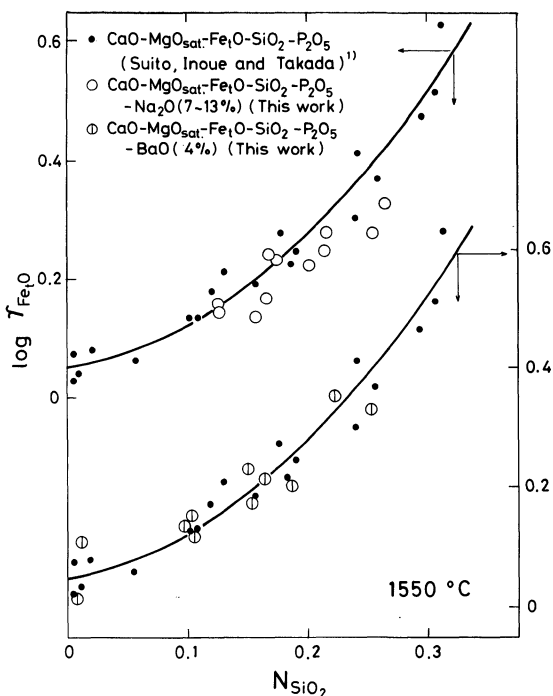
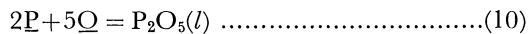


Fig. 6. Logarithms of the activity coefficients of FeO plotted against N_{SiO_2} for Na₂O- or BaO-containing and -free slags.

ship is derived.



$$\log C_P = -\Delta G^\circ/(2RT) - 1/2(\log \gamma_{\text{P}_2\text{O}_5}) + \log (\%P)/N_{\text{P}_2\text{O}_5}^{1/2} \quad \dots\dots\dots(11)$$

By substituting $\Delta G^\circ = -168\,600 + 133.0T$ (cal)¹⁴ and the $\log \gamma_{\text{P}_2\text{O}_5}$ term given by Eq. (6) into Eq. (11), the $\log C_P$ term can be expressed as a function of slag compositions and temperatures.

$$\begin{aligned} \log C_P = & 0.51(23N_{\text{CaO}} + 17N_{\text{MgO}} + 8N_{\text{Fe}_2\text{O}_3} + 33N_{\text{Na}_2\text{O}} \\ & + 42N_{\text{BaO}} + 20N_{\text{CaF}_2} + 13N_{\text{MnO}} - 26N_{\text{P}_2\text{O}_5}) \\ & + 29\,920/T - 19.280 + \log (\%P)/N_{\text{P}_2\text{O}_5}^{1/2} \quad \dots\dots\dots(12) \end{aligned}$$

The last term, $\log (\%P)/N_{\text{P}_2\text{O}_5}^{1/2}$ in Eq. (12), is dependent on the slag compositions, in particular, on the P_2O_5 content as shown in Fig. 7. The $\log (\%P)/N_{\text{P}_2\text{O}_5}^{1/2}$ calculated from the phosphorus distribution between liquid iron and MgO-saturated CaO-Fe₂O-SiO₂-P₂O₅ slags are plotted as a function of $N_{\text{P}_2\text{O}_5}$ which is represented by the hatched area. One curve in Fig. 7 is for the CaO-Fe₂O-SiO₂-P₂O₅ slags with CaO/SiO₂=3 and Fe₂O=20 wt%; the other for the CaO-Fe₂O-CaF₂-P₂O₅ slags with CaO/CaF₂=1 and Fe₂O=20 wt%. The values for $\log (\%P)/N_{\text{P}_2\text{O}_5}^{1/2}$ for the above-mentioned slags are represented by

$$\begin{aligned} \log (\%P)/N_{\text{P}_2\text{O}_5}^{1/2} = & 0.478 \log N_{\text{P}_2\text{O}_5} \\ & + 1.955 (r = 0.9996) \end{aligned}$$

and

$$\begin{aligned} \log (\%P)/N_{\text{P}_2\text{O}_5}^{1/2} = & 0.479 \log N_{\text{P}_2\text{O}_5} \\ & + 1.813 (r = 0.9999), \end{aligned}$$

respectively. One may see from Fig. 7 that the last

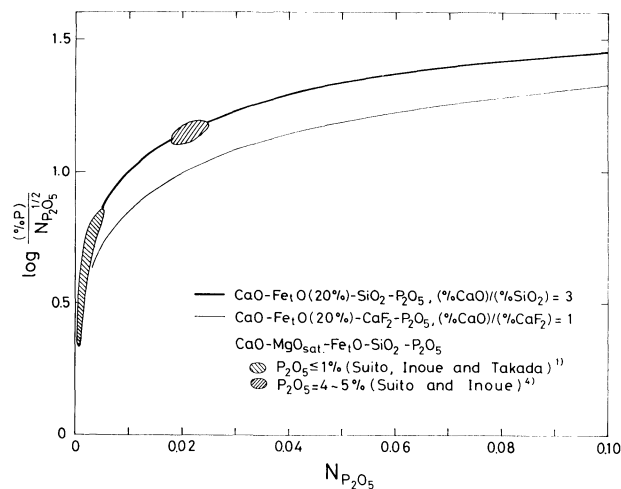


Fig. 7. Logarithms of $(\%P)/N_{\text{P}_2\text{O}_5}^{1/2}$ plotted against $N_{\text{P}_2\text{O}_5}$.

term in Eq. (12) can be considered as nearly constant under the present slag compositions at a constant P_2O_5 content. This indicates a good linear relationship in the plot of $\log C_P$ against the first term of the right hand side of Eq. (12), judging from the good linearity observed with respect to $\log \gamma_{\text{P}_2\text{O}_5}$ as shown in Fig. 5. In fact we found that good linear relationship exists between the $\log C_P$ calculated from the previous and present data and the first term in the right hand side of Eq. (12).

5. Relationship between Optical Basicity and Dephosphorization Equilibrium

Duffy and Ingram¹⁶ defined the optical basicity (theoretical) A_{th} as follows:

$$A_{\text{th}} = X_A/f_A + X_B/f_B + \dots\dots\dots(13)$$

where, X_i : the equivalent cationic fraction differing from the term (N'_i) used in the Flood-Grjotheim's treatment (Section IV. 6)

f_i : the basicity moderating parameter.

It is experimentally verified in the molten salt melts that the term f_i can be expressed by use of the Pauling's electronegativity, x_i , as follows:

$$f_i = 1.36(x_i - 0.26) \quad \dots\dots\dots(14)$$

It is found by Duffy and Ingram¹⁶ that logarithms of sulfide capacity are linearly correlated to the optical basicity. Recently, applying the optical basicity to the dephosphorization equilibrium¹⁾ obtained by the present authors, Gaskell⁵ found the term of the left hand side of Eq. (15) to be proportional to A_{th} .

$$\log N_{\text{PO}_4^{3-}}/([P][O]^{5/2}) \propto A_{\text{th}} \quad \dots\dots\dots(15)$$

where, $N_{\text{PO}_4^{3-}}$: the anionic fraction of PO_4^{3-} ion

[]: weight percent in metal phase.

The values for $\log N_{\text{PO}_4^{3-}}/([P][O]^{5/2})$ calculated from the previous data¹⁾ for the CaO-MgO_{sat}-Fe₂O-SiO₂-P₂O₅(≤1wt%) slags were plotted against the optical basicity for respective temperatures. Then, by use of the average slopes and intercepts for the corresponding temperatures, the following expression was

obtained.

$$\log N_{\text{PO}_4^{3-}}/([P][O]^{5/2}) = 22.22A_{\text{th}} + 30\,890/T - 28.752 \dots\dots\dots(16)$$

The values for $\log N_{\text{PO}_4^{3-}}/([P][O]^{5/2})$ for the slags containing Na_2O or BaO are plotted against the optical basicity as open circles and triangles in Fig. 8. These points are higher than those calculated from the previous slags without Na_2O and BaO at $1\,550^\circ\text{C}$, which are represented by solid circles. This behavior is particularly pronounced in the case of Na_2O -containing slags. These values obtained for the $\text{CaO-MgO}_{\text{sat.}}\text{-Fe}_t\text{O-SiO}_2$ slags containing $4\sim 5\text{ wt}\% \text{ P}_2\text{O}_5$ ⁴⁾ are also indicated in Fig. 8 by the symbol $\odot$. These values are also higher than those calculated from the previous slags without Na_2O and BaO at $1\,600^\circ\text{C}$. The reasons why the values for $\log N_{\text{PO}_4^{3-}}/([P][O]^{5/2})$ tend to be higher with additions of Na_2O , BaO and P_2O_5 are not determined at present.

Figure 9 shows the relationship between $\log (\%P)/N_{\text{PO}_4^{3-}}$ and $(\text{wt}\% \text{ P}_2\text{O}_5)$ for the $\text{CaO-MgO}_{\text{sat.}}\text{-Fe}_t\text{O-SiO}_2$ slags containing less than $1\text{ wt}\% \text{ P}_2\text{O}_5$ (open circle) and $4\sim 5\text{ wt}\% \text{ P}_2\text{O}_5$ (triangles), respectively. The relationships obtained in the fixed slag compositions are also illustrated in Fig. 9. If the value of $\log (\%P)/N_{\text{PO}_4^{3-}}$ is considered to be constant regardless of the slag compositions, one may predict from Eqs. (7) and (16) that $\log C_P$ can be expressed by a linear function of the optical basicity at a constant temperature.

The relationship between $\log C_P$ and the optical basicity are shown in Fig. 10 where the line was obtained by the method of linear regression for the results in the $\text{CaO-MgO}_{\text{sat.}}\text{-Fe}_t\text{O-SiO}_2\text{-P}_2\text{O}_5$ slags¹⁾ at $1\,550^\circ\text{C}$. The values for $\log C_P$ calculated from the data for the slags containing Na_2O , BaO or MnO at $1\,550^\circ\text{C}$ are plotted by the symbols $\circ$, Δ , $\square$, respectively. The A_{th} values for the pure oxides, P_2O_5 , SiO_2 , FeO , MnO , MgO , CaO , Na_2O and BaO

can be evaluated from Eqs. (13) and (14). These values are indicated by arrows in Fig. 10. The $\log C_P$ values corresponding to the pure oxides can be estimated from Eq. (12) in which the last term of the right hand side can be evaluated assuming $\text{MO-1wt}\% \text{ P}_2\text{O}_5$ slag. These results are shown in Fig. 10 by solid circles. The values of $\log C_P$ corresponding to pure FeO , MnO , MgO , CaO and Na_2O oxides are well on the line, while those for P_2O_5 , SiO_2 and BaO oxides deviate from the line as is plotted by the broken line.

6. Flood-Grjotheim's Ionic Treatment

By use of the data from the phosphorus distribution between liquid iron and $\text{CaO-MgO}_{\text{sat.}}\text{-Fe}_t\text{O-SiO}_2\text{-P}_2\text{O}_5 (\leq 1\text{ wt}\%)$ slags, the present authors demonstrated that the logarithms of the 'ideal' anionic equilibrium constant $K_{\text{PO}_4^{3-}}^{\text{id}}$ for the reaction: $2\text{P} + 5\text{O} + 3\text{O}^{2-} = 2\text{PO}_4^{3-}$ can be represented by a linear function of the equivalent cationic fractions $N'_{\text{M}^{2+}}$ as follows¹⁾;

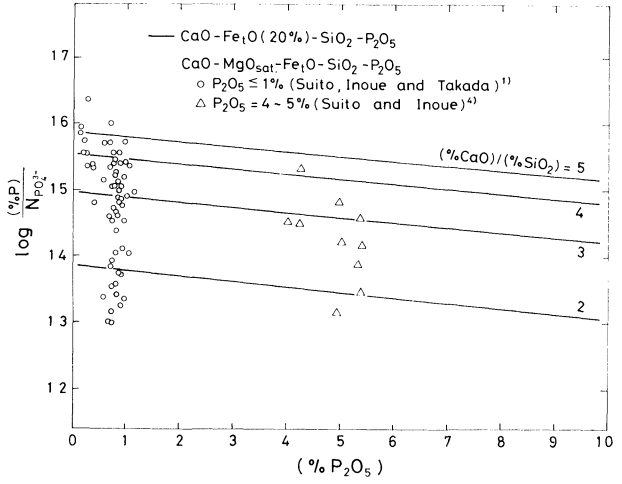


Fig. 9. Logarithms of $(\%P)/N_{\text{PO}_4^{3-}}$ plotted against $(\% \text{ P}_2\text{O}_5)$.

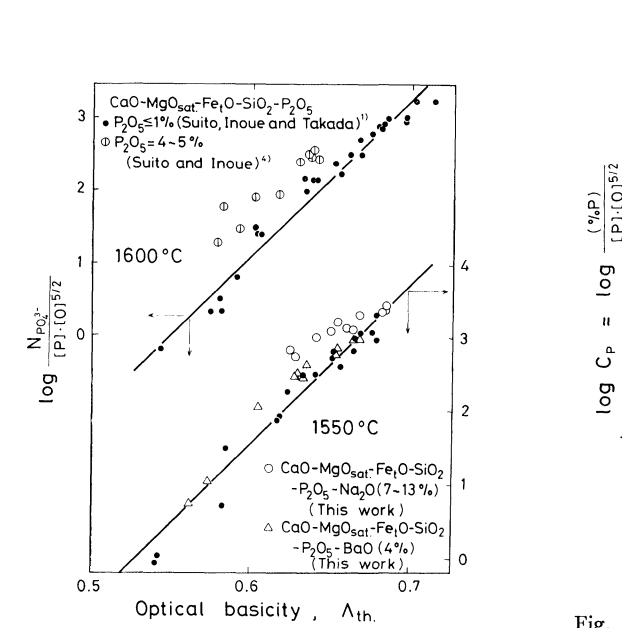


Fig. 8. Logarithms of $N_{\text{PO}_4^{3-}}/([P][O]^{5/2})$ plotted against optical basicity, A_{th} .¹⁶⁾

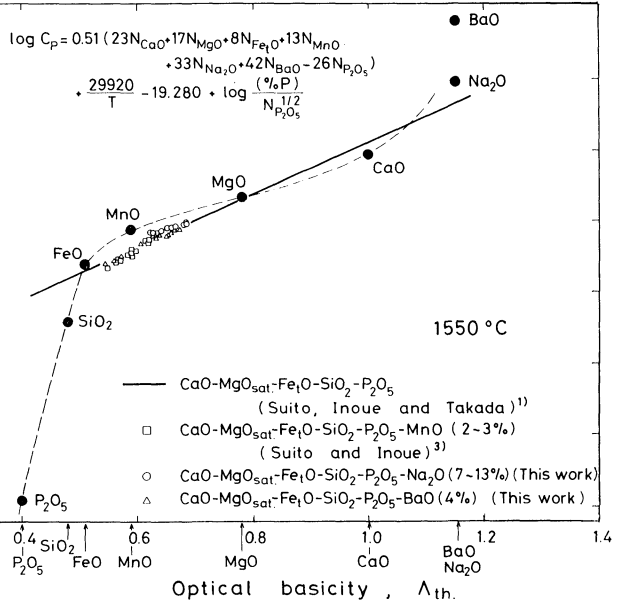


Fig. 10. Logarithms of phosphate capacity, C_P , plotted against optical basicity, A_{th} , for pure oxide, Na_2O -, BaO - or MnO -containing and -free slags.

$$\log N_{\text{PO}_4^{3-}}^2 / ([\text{P}]^2 [\text{O}]^5 N_{\text{O}^{2-}}^3) = \log K_{\text{PO}_4^{3-}}^{id} - \alpha_{\text{Ca}^{2+}} N'_{\text{Ca}^{2+}} + \alpha_{\text{Mg}^{2+}} N'_{\text{Mg}^{2+}} + \alpha_{\text{Fe}^{2+}} N'_{\text{Fe}^{2+}} \dots (17)$$

By use of the data for the previous CaO–MgO_{sat.}–Fe_tO–SiO₂–P₂O₅ (4~5wt%) slags⁴⁾ and CaO–MgO_{sat.}–Fe_tO–SiO₂–P₂O₅–MnO slags,³⁾ the effect of P₂O₅ or MnO addition on $\log K_{\text{PO}_4^{3-}}^{id}$ was studied. Consequently, the effect of the P₂O₅ content on the term in the left hand side of Eq. (17) was found to be negligible, while the parameter $\alpha_{\text{Mn}^{2+}}$ obtained from the same method described previously²⁾ was found to be about 0.6 with considerable scattering.

The values for the $\log K_{\text{PO}_4^{3-}}^{id}$ in the Na₂O- or BaO-containing slags at 1550 °C are plotted against the $(12.4N'_{\text{Ca}^{2+}} + 5.36N'_{\text{Mg}^{2+}} + 1.47N'_{\text{Fe}^{2+}})$ term which was obtained in the previous experiments¹⁾ by the method of multiple regression. These points are shown in Fig. 11 by solid circles and triangles. The results for the previous experiments of the CaO–MgO–Fe_tO–SiO₂–P₂O₅ (≤1wt%) slags at 1550 °C are also shown by small solid circles in Fig. 11 together with the line of

$$\log K_{\text{PO}_4^{3-}}^{id} / (12.4N'_{\text{Ca}^{2+}} + 5.36N'_{\text{Mg}^{2+}} + 1.47N'_{\text{Fe}^{2+}}) = 1.$$

The values of the $\alpha_{\text{Na}^{+}}$ and $\alpha_{\text{Ba}^{2+}}$ coefficients of the right hand side of Eq. (17) are determined to be 19.1 ± 3.1 and 24.0 ± 12.6 , respectively by the same method described previously.²⁾ The values for $\log K_{\text{PO}_4^{3-}}^{id}$ calculated from the experiments of Na₂O- and BaO-containing slags are plotted against the $(12.4N'_{\text{Ca}^{2+}} + 5.36N'_{\text{Mg}^{2+}} + 1.47N'_{\text{Fe}^{2+}} + 19.1N'_{\text{Na}^{+}} + 24.0N'_{\text{Ba}^{2+}})$ term as open circles and triangles, respectively. It is seen that these points lie on the line within the scattering of the experimental points given by small solid circles.

V. Summary

The results obtained are summarized in the following.

(1) The effect of Na₂O or BaO addition to the CaO–MgO–Fe_tO–SiO₂–P₂O₅ slags on the logarithms of equilibrium quotient $(\% \text{P}_2\text{O}_5) / ([\% \text{P}]^2 \cdot [\% \text{Fe}_t\text{O}]^5)$, which was expressed by a linear function of the slag compositions (in wt%), was investigated. The CaO-equivalencies of Na₂O and BaO were observed to be 1.2 and 0.9 on weight percent basis, respectively.

(2) The effect of Na₂O or BaO addition on the logarithms of the activity coefficient of P₂O₅ was studied using Turkdogan and Pearson's treatment. The CaO-equivalencies of Na₂O and BaO were observed to be 33/23 and 42/23 on mole fraction basis, respectively.

(3) Activity coefficients of Fe_tO in the Na₂O- or BaO-containing slags are compared with those in the slags without Na₂O and BaO in the plot of $\log \gamma_{\text{Fe}_t\text{O}} \text{ vs. } N_{\text{SiO}_2}$. It was found that the Na₂O addition decreases activity coefficients of Fe_tO, whereas the BaO addition has no effect.

(4) The values for $\ln N_{\text{PO}_4^{3-}} / ([\text{P}][\text{O}]^{5/2})$, were found to be linearly related to the theoretical optical basicity

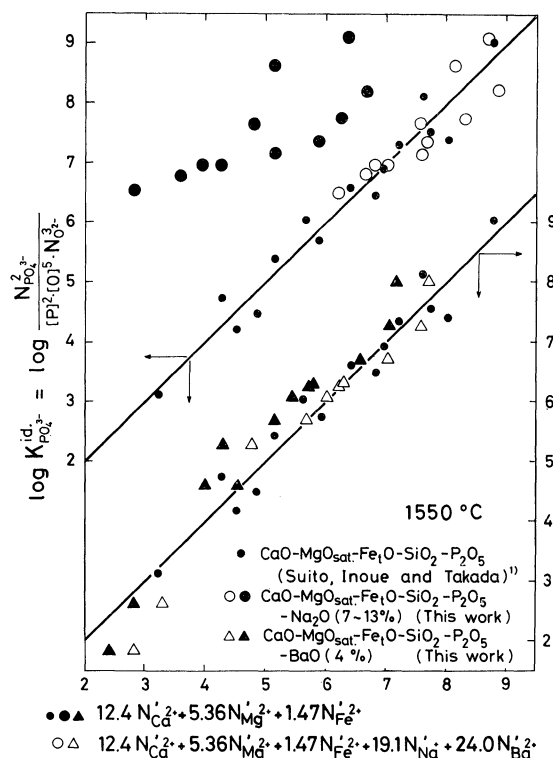


Fig. 11. Ionic treatment between $\log K_{\text{PO}_4^{3-}}^{id}$ and $\alpha_i N'_i$ for Na₂O- or BaO-containing and -free slags.

in the Na₂O- or BaO-containing slags as well as in the slags without Na₂O and BaO. The values for $\ln N_{\text{PO}_4^{3-}} / ([\text{P}][\text{O}]^{5/2})$ at a constant A_{th} value, however, tend to increase with the addition of Na₂O or BaO.

(5) The values of coefficients of the equivalent cationic fraction, $N'_{\text{Mn}^{2+}}$, $N'_{\text{Na}^{+}}$ and $N'_{\text{Ba}^{2+}}$ in the Flood–Grjotheim ionic equation are determined as 0.6, 19 and 24, respectively.

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